



## 光影景觀中多維度時間的敘事設計方法

Multidimensional Temporal Narratives in  
Light-and-Shadow Landscapes

### Keyword

+Capturing  
+Clarifying  
+Translating

Liao

Yi-Chun

## 摘要-Summary

本設計研究以「光學三要素」——光源、不透明障礙物、受光面——為核心架構，探討淡水河岸景觀中多維度時間的敘事可能。透過 05:00-23:00 的長時間序列觀測與 10 秒動態影像紀錄，本計畫釐清了欄杆（構造物）、波光（動態光源）與影子（顯影紀錄）三者間的聯動關係。研究發現，時間感不僅存在於太陽的位移，更體現在人為活動與光影節奏的「對位」之中。本設計旨在透過構築操縱，將河岸轉化為一個感知瞬時、循環與永恆時間的敘事容器。

This design research is structured around the "Three Optical Elements"—Light Source, Opaque Obstacle, and Receiving Surface—to explore the narrative potential of multi-dimensional time within the Tamsui riverbank landscape. Through a long-term sequential observation (05:00-23:00) and 10-second dynamic video recordings, this project clarifies the interlocking relationship between railings (tectonic structures), water shimmer (dynamic light source), and shadows (developing records). The study reveals that the sense of time exists not only in the sun's displacement but also in the "alignment" between human activity and light-shadow rhythms. The design aims to transform the riverbank into a narrative vessel for perceiving momentary, cyclic, and eternal time through tectonic manipulation.

## 設計操作說明

首階段透過 05:00-23:00 的長時序監測，每隔兩小時進行靜態擷取與 10 秒動態影像紀錄，將天文位移與水面波光轉化為可分析的數位資料庫。接著，透過標註「有人時段」與「影子時段」進行行為對位，釐清欄杆構件如何作為時間刻度，以及波光頻率如何誘發人群的停留凝視。

最終轉化階段，分析結果落實於界面設計：透過操縱不透明障礙物的幾何孔隙進行「幾何裁切」，並微調受光面的材質反射率與坡度，使其成為捕捉瞬時波光與永恆影跡的「顯影介面」。此操作不僅精準編排了光影在空間中的律動，更將物理光學轉化為具備行為共鳴的景觀敘事，使淡水河岸成為一個能跨越維度、紀錄時間流變的感知容器。

The initial phase involves a long-term sequential monitoring from 05:00 to 23:00, capturing static frames every two hours alongside 10-second dynamic video clips. This process transforms astronomical displacement and water shimmer into an analytical digital database. Subsequently, by mapping "occupied periods" against "shadow periods," a behavioral alignment is established to clarify how railing components function as temporal scales and how the frequency of water shimmer induces human lingering and contemplation.

In the final transformation phase, these analytical findings are integrated into the interface design. Through the "geometric clipping" of opaque obstacles' porosity and the fine-tuning of the receiving surfaces' material reflectivity and slope, the site becomes a "developing interface" that captures both instantaneous shimmer and eternal shadows. This operation not only precisely choreographs the rhythm of light and shadow within the space but also translates physical optics into a landscape narrative that resonates with human behavior, turning the Tamsui riverbank into a perceptive vessel that records the flux of time across multiple dimensions.